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# Machin's Flagellar Wave Attenuation Revisited via a Lattice Boltzmann–Immersed Boundary Method

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**Abstract**

Flagellar locomotion at low Reynolds number results from the interaction between elasticity, viscous dissipation, and hydrodynamic forces. This work presents a two-stage computational framework combining the classical finite-length Machin model with a lattice Boltzmann–immersed boundary (LBM–IBM) formulation. Model I solves the passive Machin boundary-value problem for $Sp = 5$, yielding the reference amplitude and phase along the filament. The distal-to-proximal amplitude ratio is 0.2063812033, corresponding to a 79.36% reduction. In Model II, this deformation is prescribed in a D2Q9 LBM coupled to a direct-forcing IBM and simulated for five forcing periods. Density variations remain small and the flow remains in a low-Mach-number regime. The two models form the first stages of a broader hierarchy. Model III is proposed as a passive LBM–IBM fluid–structure interaction model in which deformation becomes an unknown, while Model IV introduces a phenomenological active bending moment. The framework therefore provides a controlled route from prescribed elastohydrodynamic kinematics to future passive and active fluid–structure interaction.

**Keywords**: Flagellar dynamics, Machin model, lattice Boltzmann method, immersed boundary method, elastohydrodynamics

## 1 INTRODUCTION

Flagellar locomotion is a classical problem in low-Reynolds-number hydrodynamics involving elasticity, viscous dissipation, and propulsion. Spermatozoa and other microswimmers move through the deformation of slender flexible filaments, for which inertial effects are generally small (Gray and Hancock, 1955; Lauga and Powers, 2009).

A fundamental description of flagellar wave propagation was introduced by Machin (Machin, 1958). His passive elastohydrodynamic formulation describes the competition

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between bending elasticity and viscous drag, predicting spatial attenuation and phase propagation of harmonic deformations. Wiggins and Goldstein (Wiggins and Goldstein, 1998) established the sperm number as a relevant dimensionless parameter for flexible filaments, while subsequent studies investigated driven filaments and microorganism locomotion (Wiggins et al., 1998; Lauga and Powers, 2009).

Mathematical models have also been applied to spermatozoa in confined geometries (Montenegro-Johnson et al., 2015) and to the broader mathematical description of sperm motility (Gaffney et al., 2021). For numerical fluid–structure interaction, the immersed boundary method (IBM) is particularly suitable for slender moving structures because the fluid can be solved on a fixed Cartesian grid while the filament is represented by Lagrangian points (Peskin, 2002). Direct-forcing formulations have been developed for complex moving boundaries and particulate flows (Fadlun et al., 2000; Kim et al., 2001; Balaras, 2004; Uhlmann, 2005; Taira and Colonius, 2007; Pinelli et al., 2010; Kempe and Fröhlich, 2012; Kim et al., 2019).

The combination of LBM and IBM provides a convenient framework for low-Reynolds-number flows with moving interfaces. The lattice Boltzmann method supplies the Eulerian fluid solver, while the IBM transfers velocity and force information between Eulerian and Lagrangian descriptions (Succi, 2001; Peskin, 2002). The forcing formulation of Guo, Zheng and Shi (Guo et al., 2002) is adopted here. Explicit direct-forcing IBM may exhibit boundary slip because of the non-reciprocity of interpolation and spreading; Gsell and Favier (Gsell and Favier, 2021) proposed a correction for this issue. The present implementation does not include that correction.

The objective of this work is to establish a controlled computational hierarchy rather than a complete active spermatozoon model. The hierarchy consists of **Model I**, the classical finite-length passive Machin boundary-value problem; **Model II**, prescribed Machin kinematics coupled to LBM–IBM; **Model III**, a future passive LBM–IBM fluid–structure interaction model; and **Model IV**, a future active LBM–IBM fluid–structure interaction model. Only Models I and II are implemented here. In Models I–II the filament deformation is prescribed, whereas in Models III–IV it becomes an unknown of the coupled problem.

## 2 METHODOLOGY

### *2.1 Model I: finite-length Machin problem*

Let $h(s,t)$ denote the transverse displacement of a slender filament, where $s$ is the arclength and $t$ is time. For compactness, $h^{(n)} = \partial^n h/\partial s^n$, $n = 1,\dots,4$. The passive linear elastohydrodynamic balance is

$$\zeta_\perp h_t = -A h^{(4)}, \tag{1}$$

where $A$ is the bending stiffness and $\zeta_\perp$ is the transverse viscous drag coefficient.

For harmonic motion, the characteristic elastohydrodynamic length is $\ell = (A/\zeta_\perp\omega)^{1/4}$ and the sperm number is $Sp = L/\ell$. The displacement is written as $h(s,t) = h_0\,\mathrm{Re}\{F(\chi)e^{i\omega t}\}$, where $\chi = s/\ell$. The resulting dimensionless problem is

$$F^{(4)} + iF = 0, \tag{2}$$

with $F(0) = 1$, $F^{(2)}(0) = 0$, $F^{(2)}(\Lambda) = 0$, and $F^{(3)}(\Lambda) = 0$, where $\Lambda = L/\ell = Sp$.

The dimensionless amplitude and phase are $\mathcal{A} = |F|$ and $\phi = \arg(F)$. The attenuation ratio is $R_A = \mathcal{A}(\Lambda)/\mathcal{A}(0)$. For the reference case $\Lambda = Sp = 5$, the numerical solution gives $R_A = 0.2063812033$, corresponding to a 79.36% reduction of the wave amplitude along the finite filament. The dominant decaying mode of the corresponding semi-infinite problem is $F(\chi) \sim \exp[-\cos(\pi/8)\chi + i\sin(\pi/8)\chi]$.

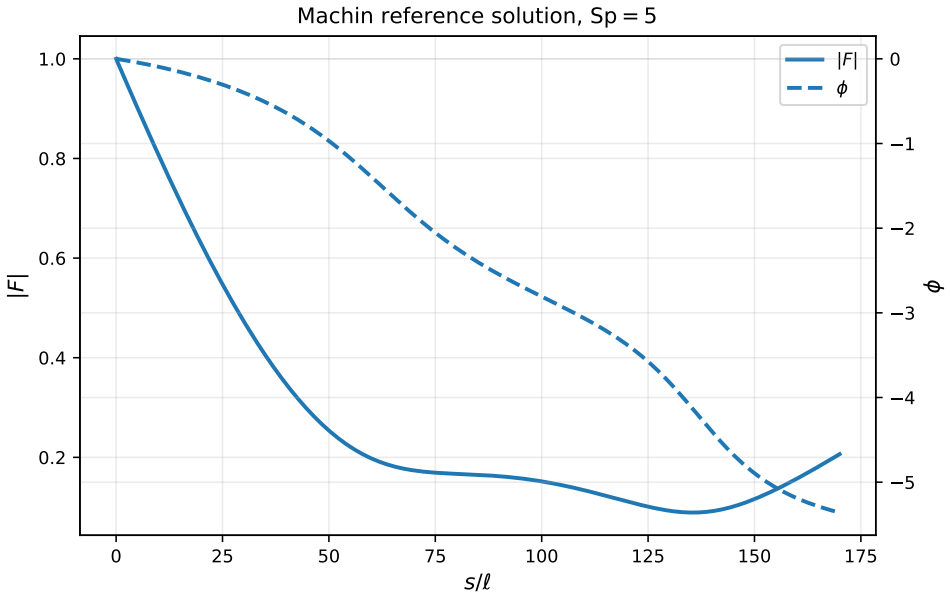



**Fig. 1:** Dimensionless amplitude and phase distributions obtained from the finite-length Machin boundary-value problem for $Sp = 5$.

### *2.2 Model II: prescribed Machin kinematics with LBM–IBM*

The second stage couples the Machin deformation to a D2Q9 lattice Boltzmann solver with BGK collision, Guo forcing, and a direct-forcing immersed boundary method. The lattice populations evolve according to

$$f_i(\boldsymbol{x} + \boldsymbol{c}_i\Delta t, t + \Delta t) = f_i - \frac{1}{\tau}(f_i - f_i^{eq}) + \Delta t\,\Phi_i, \tag{3}$$

where the equilibrium distribution is

$$f_i^{eq} = w_i\rho\left[1 + \frac{\boldsymbol{c}_i\cdot\boldsymbol{u}}{c_s^2} + \frac{(\boldsymbol{c}_i\cdot\boldsymbol{u})^2}{2c_s^4} - \frac{\boldsymbol{u}\cdot\boldsymbol{u}}{2c_s^2}\right]. \tag{4}$$

The macroscopic density is $\rho = \sum_i f_i$, the velocity is $\boldsymbol{u} = \sum_i f_i\boldsymbol{c}_i/\rho + \boldsymbol{F}/(2\rho)$, and the kinematic viscosity is $\nu = c_s^2(\tau - 1/2)\Delta t$.

The filament is represented by Lagrangian points $\boldsymbol{X}_j = (x_j, y_j)$. Its prescribed deformation is $h(s,t) = h_0\,\mathrm{Re}\{F(\chi)e^{i\omega t}\}$, with $\chi = \Lambda s/L_f$. Thus $x_j = x_{j,0}$, $y_j = y_c + h(s_j,t)$, $v_x = 0$, and $v_y = h_0\,\mathrm{Re}\{i\omega F(\chi)e^{i\omega t}\}$.

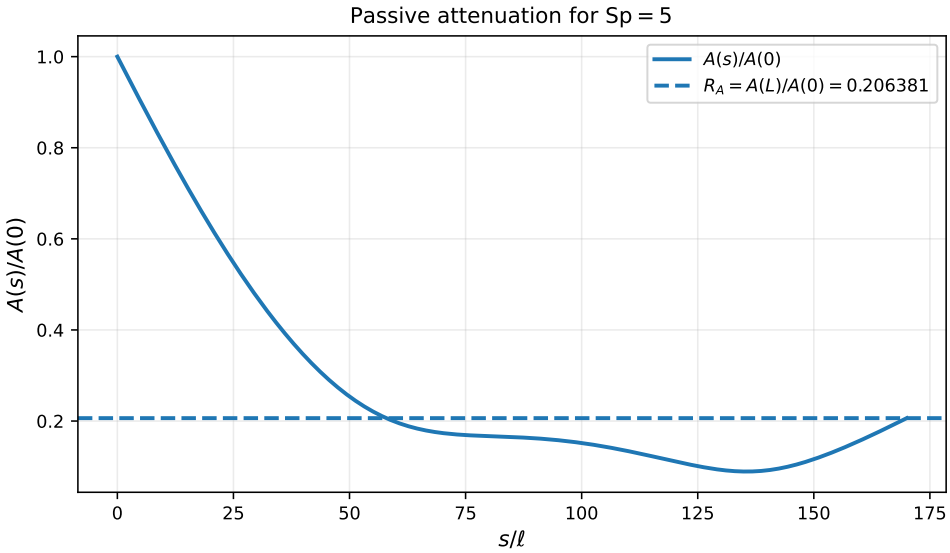



**Fig. 2:** Wave-amplitude attenuation along the finite filament for the reference Machin solution.

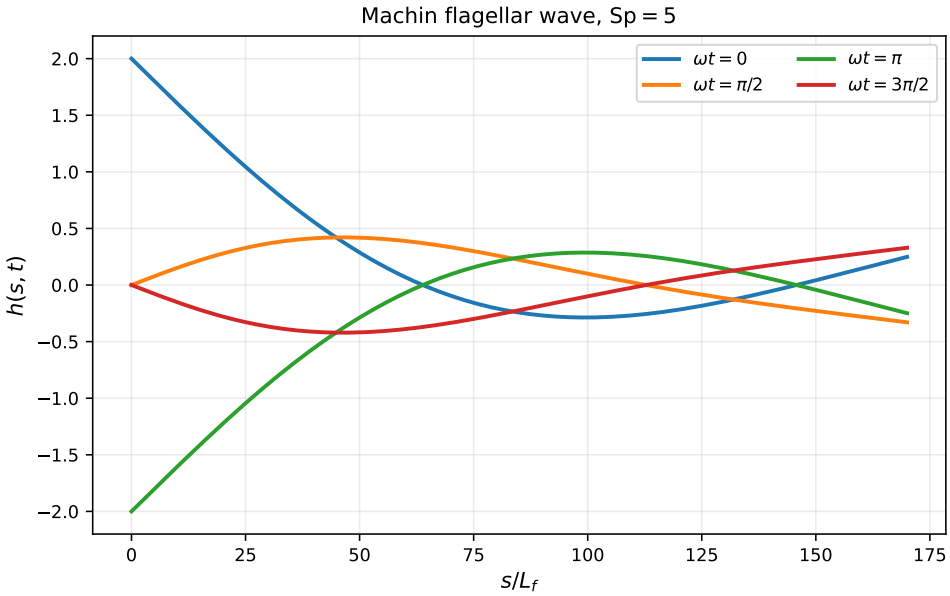



**Fig. 3:** Representative flagellar configurations reconstructed from the Machin solution over one oscillation period.

Using a regularized four-point Peskin kernel, the interpolated velocity is $\boldsymbol{U}_j = \sum_{\boldsymbol{x}} \boldsymbol{u}(\boldsymbol{x})\delta_h(\boldsymbol{x}-\boldsymbol{X}_j)\Delta x^2$. The direct-forcing force is $\boldsymbol{f}_j = \rho_0(\boldsymbol{U}_j^b - \boldsymbol{U}_j)/\Delta t$, while the Eulerian immersed-boundary force is $\boldsymbol{F}_{IB} = \sum_j \boldsymbol{f}_j \delta_h(\boldsymbol{x} - \boldsymbol{X}_j)$.

The filament deformation remains prescribed throughout Model II and is not modified by the hydrodynamic force. This stage is therefore a hydrodynamic benchmark for prescribed kinematics rather than a complete fluid–structure interaction model. The explicit direct-forcing implementation does not include the correction proposed by Gsell and Favier (Gsell and Favier, 2021); local slip or penetration errors may therefore remain.

### *2.3 Numerical parameters*

The Eulerian domain contains $300 \times 120$ nodes, with $\Delta x = \Delta t = 1$, $\rho_0 = 1$, $\nu = 0.08$, and $\tau = 0.74$. The numerical filament has $L_f = 170$ and 180 Lagrangian points, starting at $x_0 = 30$, with amplitude $h_0 = 2$. The angular frequency is $\omega = 2\pi/1600$, giving a forcing period $T = 1600$. The simulation runs until $t = 8000$, corresponding to five periods, with $\Lambda = Sp = 5$.

## 3 RESULTS AND DISCUSSION

### *3.1 Machin reference solution*

The numerical solution exhibits the expected amplitude attenuation and phase variation along the filament, as shown in Fig. 1. For $Sp = 5$, the distal amplitude is 20.64% of the proximal amplitude, as confirmed by Fig. 2. The reconstructed configurations in Fig. 3 provide the prescribed kinematics for Model II.

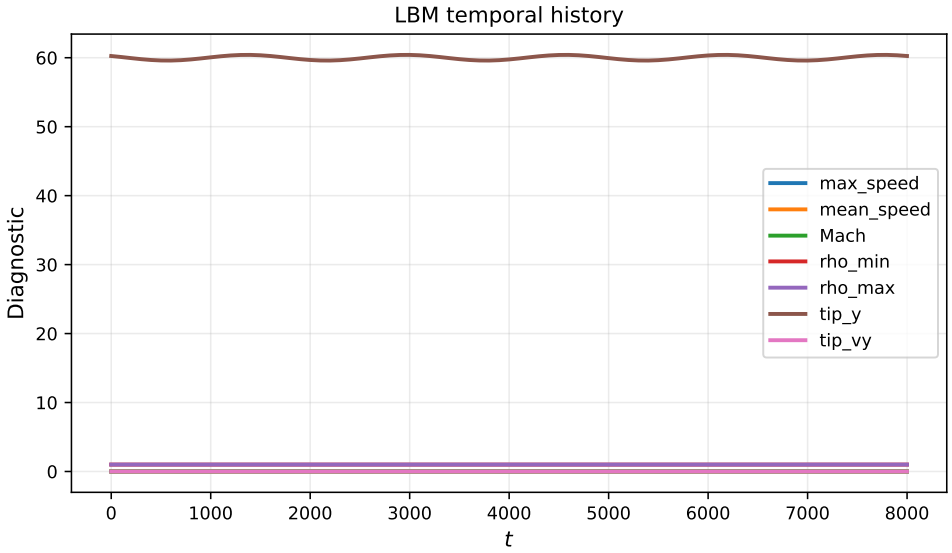



**Fig. 4:** Temporal history of the principal quantities monitored during the LBM–IBM simulation.

### *3.2 LBM–IBM hydrodynamic response*

The prescribed Machin waveform was propagated through the fluid for five periods. Figure 4 shows the temporal evolution of the monitored hydrodynamic quantities and indicates stable behavior throughout the simulation.

The tip displacement and velocity are shown in Figs. 5 and 6. Since the filament deformation is prescribed, these quantities represent the transfer of the Machin kinematics to the immersed boundary rather than a structural response obtained from a coupled equilibrium.

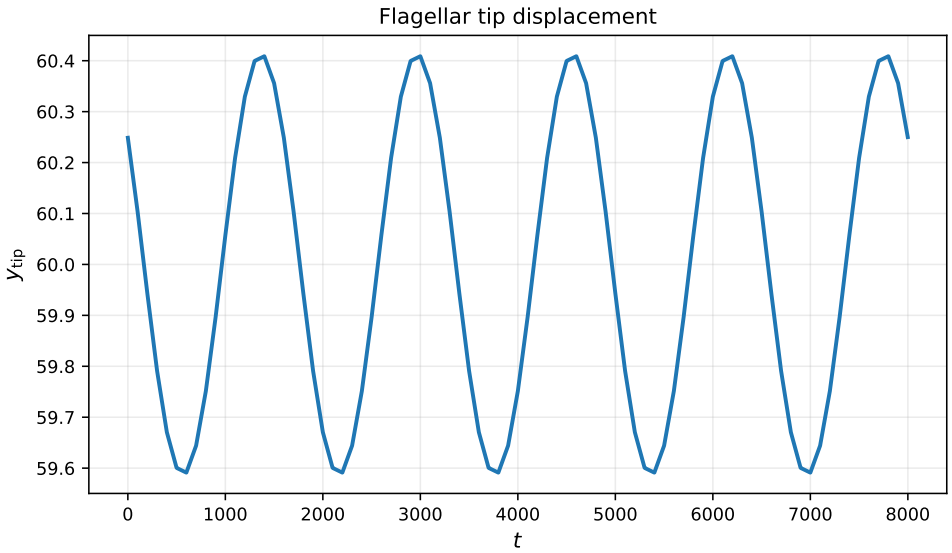



**Fig. 5:** Prescribed transverse displacement of the filament tip during the LBM–IBM simulation.

The density field remains close to the reference state (Fig. 7), with $\rho_{\min} = 0.998627124$ and $\rho_{\max} = 1.001341410$. The maximum velocity components are approximately $\max|u_x| = 3.00 \times 10^{-3}$ and $\max|u_y| = 2.79 \times 10^{-3}$. The maximum force magnitude is approximately $1.40 \times 10^{-3}$, with $\max|F_y| \approx 1.33 \times 10^{-3}$.

These values indicate small density fluctuations and a low-Mach-number flow. The simulation therefore provides a stable hydrodynamic realization of the prescribed Machin deformation.

The main physical result is the attenuation predicted by Model I, $R_A = 0.2063812033$, while the main numerical result of Model II is the stable transfer of this prescribed deformation to the LBM–IBM fluid. Model II should therefore not be interpreted as structural validation, because the filament waveform is not determined by the hydrodynamic force.

## 4 PERSPECTIVES: PASSIVE AND ACTIVE FSI

The present models constitute the first two stages of a four-model hierarchy. The next stages remove the prescribed-kinematics assumption and introduce bidirectional fluid–structure interaction.

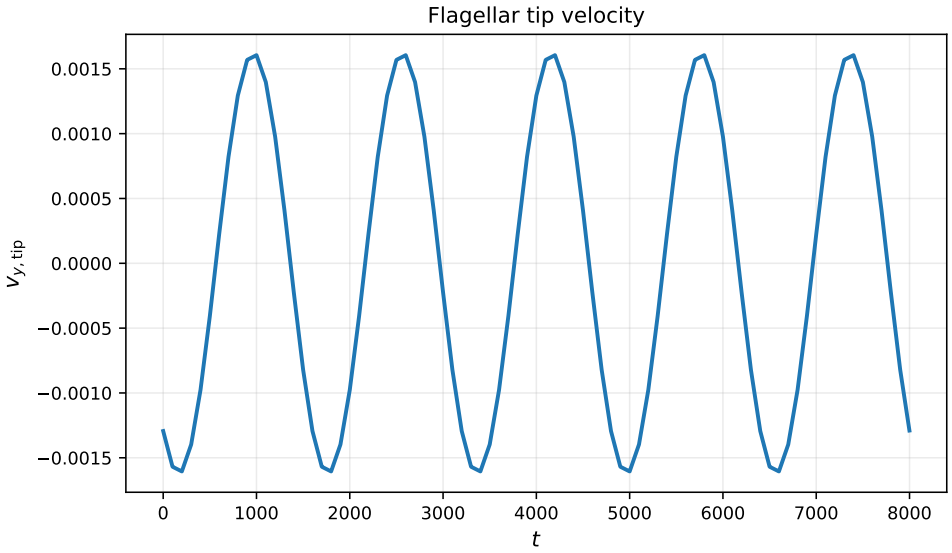



**Fig. 6:** Prescribed transverse velocity of the filament tip during the five forcing periods.

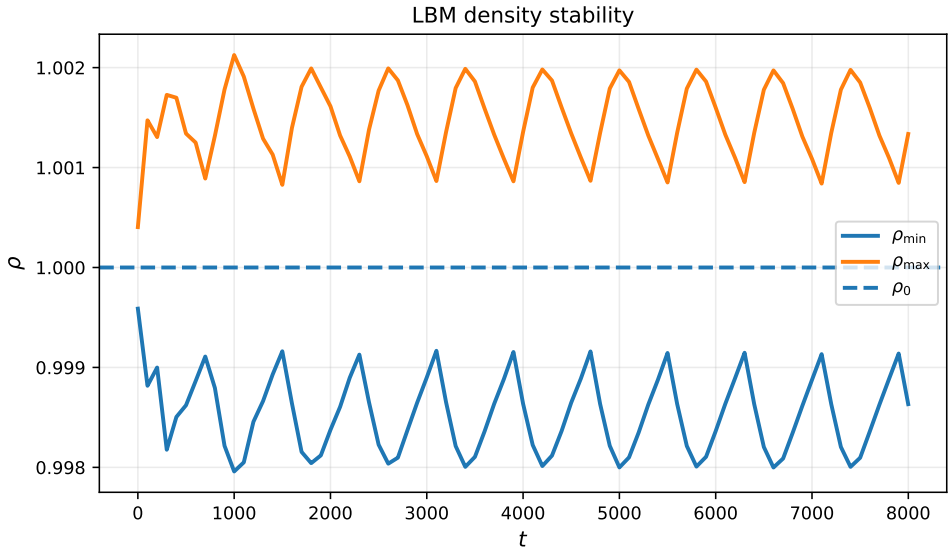



**Fig. 7:** Density field obtained from the LBM–IBM simulation at the reported final state.

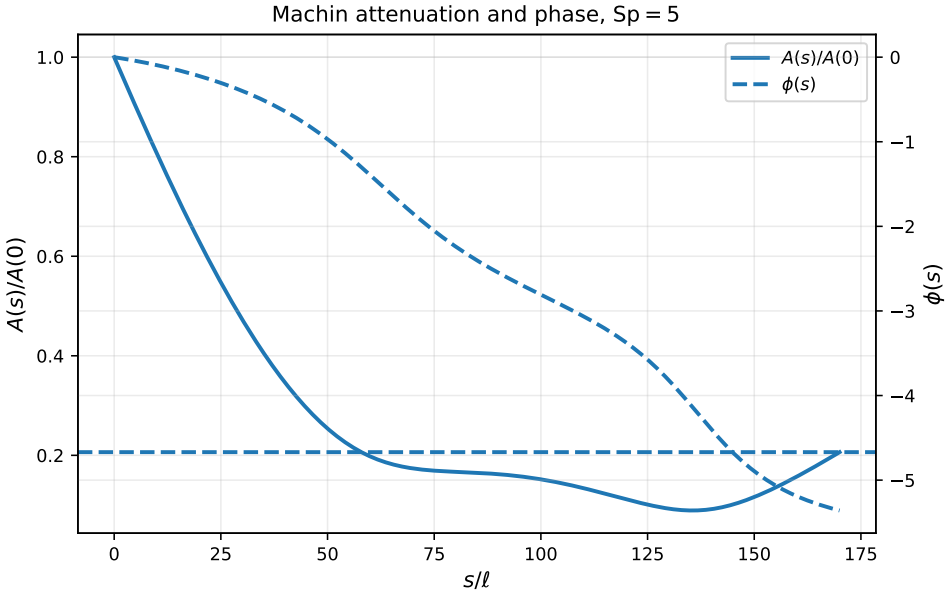



**Fig. 8:** Summary of the principal quantities obtained from the finite-length Machin reference solution.

### *4.1 Model III: passive LBM–IBM FSI*

In Model III, the filament deformation becomes an unknown and responds to the hydrodynamic force generated by the fluid. The proposed passive structural evolution is

$$\frac{h^{n+1} - h^n}{\Delta t} = \frac{1}{\zeta_s}\left[-AD_4 h^{n+1} + f_h^n\right]. \tag{5}$$

Equivalently, $[I + (\Delta t A/\zeta_s)D_4]\, h^{n+1} = h^n + (\Delta t/\zeta_s) f_h^n$. The proposed boundary conditions are $h(0) = 0$, $h^{(1)}(0) = 0$, $h^{(2)}(L) = 0$, and $h^{(3)}(L) = 0$.

Here, $f_h$ is the hydrodynamic reaction obtained from the IBM force. Without distributed active actuation, the passive structure is expected to relax toward a static configuration. An initial perturbation would therefore be used to investigate stability and transient response.

The relaxation procedure is a numerical mechanism for approaching quasi-static structural balance and should not be interpreted as a physical structural damping law.

### *4.2 Model IV: active LBM–IBM FSI*

Model IV introduces an internal active bending moment while retaining bidirectional coupling with the fluid. The proposed structural balance is $0 = -Ah^{(4)} + M_a^{(2)} + f_h$, together with the phenomenological law $\tau_a M_{a,t} + M_a = M_{\max}\tanh(\kappa/\kappa_c)$ and $\kappa \approx h^{(2)}$.

The proximal conditions are $h(0) = 0$ and $h^{(1)}(0) = 0$, while the distal conditions are $-Ah^{(2)}(L) + M_a(L) = 0$ and $-Ah^{(3)}(L) + M_a^{(1)}(L) = 0$.

No travelling-wave waveform is prescribed in Model IV. Sustained oscillations or travelling waves must emerge from the coupled dynamics and be measured from the numerical solution. The active law is phenomenological and does not by itself guarantee self-excited

oscillations.

Models III and IV are future extensions and are not presented as results of the present study. They define the intended progression from prescribed Machin kinematics to passive and active fluid–structure interaction.

## 5 CONCLUSIONS

A two-stage computational framework was developed to connect the finite-length Machin model with a D2Q9 lattice Boltzmann–immersed boundary simulation of prescribed flagellar deformation.

For $Sp = 5$, the finite-length Machin problem yields a distal-to-proximal amplitude ratio of 0.2063812033, corresponding to a 79.36% attenuation. This solution supplies the reference kinematics for Model II.

The prescribed waveform was coupled to a D2Q9 BGK LBM with Guo forcing and direct-forcing IBM and simulated for five forcing periods. The density remained within $0.998627124 \leq \rho \leq 1.001341410$, while the velocity magnitudes remained small, indicating a stable low-Mach-number computation.

The main limitation is that Model II does not solve the structural problem: the deformation is imposed and therefore cannot respond to the hydrodynamic force. It should consequently be regarded as a hydrodynamic benchmark rather than a complete FSI model.

The work establishes the first two stages of a four-model hierarchy. Model III will introduce passive bidirectional LBM–IBM FSI, with the deformation determined by the coupled structural and hydrodynamic balance. Model IV will additionally introduce an active internal bending moment without prescribing a travelling waveform. These extensions provide a progressive route from classical elastohydrodynamics to passive and active computational fluid–structure interaction, with potential relevance to the modeling of microswimmers, bioinspired microrobots, and other small-scale fluidic systems.

### Code availability

The source code and input files used to generate the numerical results presented in this work are openly available at the GitHub repository: `https://github.com/LuisVCSilva/enmc-2027__machin-flagellar-lbm-ibm`.

## 6 *Acknowledgements*

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